

$$\vec{L} = \vec{r} \times \vec{p} \quad ; \quad \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{p} = p_x\hat{i} + p_y\hat{j} + p_z\hat{k}$$

$$\vec{L} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} = (yp_z - zp_y)\hat{i} -$$

$$(xp_z - zp_x)\hat{j} +$$

$$(xp_y - yp_x)\hat{k}$$

$$\vec{L} = (x\hat{i} + y\hat{j} + z\hat{k}) \times (p_x\hat{i} + p_y\hat{j} + p_z\hat{k})$$

$$[A, B] = AB - BA$$

; $\hat{x}$

OPERADOR POSICIÓN

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}$$

OPERADOR MOMENTUM

$$[x, p] = xp - px = -i\hbar x \frac{\partial}{\partial x} - \left(-i\hbar \frac{\partial}{\partial x} x \right)$$

$$= -i\hbar x \frac{\partial}{\partial x} + i\hbar \frac{\partial}{\partial x} x$$

$$\psi(x): [x, p] \psi(x) = -i\hbar x \frac{\partial}{\partial x} \psi(x) + i\hbar \frac{\partial}{\partial x} (x \cdot \psi(x))$$

$$= \cancel{-i\hbar x \frac{\partial \psi}{\partial x}} + i\hbar \cdot 1 \cdot \psi(x) + \cancel{i\hbar x \frac{\partial \psi}{\partial x}}$$

$$[x, p] \psi(x) = i\hbar \psi(x)$$

$$\Rightarrow [x, p] = i\hbar \quad (\text{RELACIÓN DE CONJUGACIÓN CAUSAL})$$

$$\hat{L} = (L_x, L_y, L_z) \quad \rightarrow \quad z p_x - x p_z = -i\hbar z \frac{\partial}{\partial x} - (-i\hbar x \frac{\partial}{\partial z})$$

$$\downarrow$$

$$y p_z - z p_y = -i\hbar y \frac{\partial}{\partial z} - (-i\hbar z \frac{\partial}{\partial y})$$

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

$$L^2 = L_+ L_- + L_- L_+ - \hbar L_z \quad \left(L_x = \frac{L_+ + L_-}{2} \right)$$

$$(L_x^2 + L_y^2 + L_z^2) \quad \left(L_y = \frac{L_+ - L_-}{2i} \right)$$

$$[L_z, L_+] = \hbar L_+$$

$$\Rightarrow L_z L_+ - L_+ L_z = \hbar L_+$$

$$L_z (L_+ |\alpha, \beta\rangle) = (\hbar L_+ + L_+ L_z) |\alpha, \beta\rangle$$

$$= \hbar L_+ |\alpha, \beta\rangle + L_+ \underbrace{L_z |\alpha, \beta\rangle}_{\hbar \beta |\alpha, \beta\rangle}$$

$$= \hbar L_+ |\alpha, \beta\rangle + \hbar \beta L_+ |\alpha, \beta\rangle$$

$$\underbrace{L_z L_+ |\alpha, \beta\rangle}_{|\phi\rangle} = \hbar(\beta+1) \underbrace{L_+ |\alpha, \beta\rangle}_{|\phi\rangle} \quad \hbar \beta |\alpha, \beta\rangle$$

$$\Rightarrow L_+ |\alpha, \beta\rangle \propto (\beta+1) |\alpha, \beta\rangle$$

$$L^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle$$

$$L_z |l, m\rangle = \hbar m |l, m\rangle$$

$$m = -l, \dots, l$$

$$\# m = 2l + 1$$

$$\text{si } l=1, \quad m = 1, 0, -1$$

$$|l, m\rangle: |1, 1\rangle ; |1, 0\rangle ; |1, -1\rangle$$

$$\Rightarrow |1, 1\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} ; |1, 0\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} ; |1, -1\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\langle i | A | i \rangle$$

$$A = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

$$|i=1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|i=2\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\langle i | A | j \rangle$$

$$A = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

$$|i=1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|i=2\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\langle 1 | A | 1 \rangle = (1 \ 0) \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= (1 \ 0) \begin{pmatrix} 2 \\ 5 \end{pmatrix} = 2$$

$$\langle 1 | A | 2 \rangle = (1 \ 0) \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (1 \ 0) \begin{pmatrix} 3 \\ 7 \end{pmatrix} = 3$$

$\Rightarrow$

$$A_{ij} = \langle i | A | j \rangle$$

$$\begin{aligned}
 \langle l, m' | L_z | l, m \rangle &= \langle l, m' | \hbar m | l, m \rangle \\
 &= \hbar m \langle l, m' | l, m \rangle = \begin{cases} \hbar m, & m = m' \\ 0, & m \neq m' \end{cases}
 \end{aligned}$$

$$\Rightarrow L_z = \hbar \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(T): $L^2 = ?$